AEIA Engineering Analysis Report

Dataset: engine_test_run.csv
Rows: 61 | Columns: 7
Session ID: S00001
Generated: 2026-07-12 21:31:58
AEIA Version: 0.1.0

Table of Contents

Statistical Summary

Column	Mean	Std Dev	Min	Max	Q1	Q3	Trend
Sample_ID	30.5000	17.4642	1.0000	60.0000	15.7500	45.2500	Increasing
Engine_Temp_C	91.7507	5.9313	86.8500	134.5000	89.7350	92.7525	Increasing
Oil_Pressure_psi	39.6514	1.6187	36.1200	43.1400	38.5375	40.8725	Stable
RPM	2495.3433	19.2550	2456.8000	2540.0000	2484.7000	2511.2500	Stable
Vibration_mm_s	0.2557	0.0565	0.1860	0.6200	0.2288	0.2690	Stable

Engineering Findings

Engine_Temp_C

[Warning] Column 'Engine_Temp_C' shows an unusual value of 134.5 at row 44, which is 7.21 standard deviations from the mean — flagged as a statistical outlier (Warning).

[Warning] Column 'Engine_Temp_C' has a value of 134.5 at row 44 that falls outside the expected range (below 85.21 or above 97.28) — flagged as an outlier (Warning).

[Info] Column 'Engine_Temp_C' shows a Increasing trend over the dataset (slope = 0.1383).

[Critical] Engine temperature exceeded the critical safety threshold of 120°C. (triggered because Engine_Temp_C = 134.5 at row 44 > 120).

[Warning] This channel shows a sustained upward drift over the recorded samples. (triggered because trend_slope(Engine_Temp_C) = 0.1383 > 0.05).

Vibration_mm_s

[Warning] Column 'Vibration_mm_s' shows an unusual value of 0.62 at row 29, which is 6.45 standard deviations from the mean — flagged as a statistical outlier (Warning).

[Warning] Column 'Vibration_mm_s' has a value of 0.62 at row 29 that falls outside the expected range (below 0.17 or above 0.33) — flagged as an outlier (Warning).

[Warning] This channel shows unusually high variability relative to its average value, indicating possible instability. (triggered because coefficient_of_variation(Vibration_mm_s) = 0.2209 > 0.15).

Sample_ID,Engine_Temp_C,Oil_Pressure_psi,RPM,Vibration_mm_s

[Warning] Row 0 was flagged as an unusual combination of values across Sample_ID, Engine_Temp_C, Oil_Pressure_psi, RPM, Vibration_mm_s — this pattern does not resemble the rest of the dataset (Warning).

Sample_ID

[Info] Column 'Sample_ID' shows a Increasing trend over the dataset (slope = 1.0000).

[Warning] This channel shows a sustained upward drift over the recorded samples. (triggered because trend_slope(Sample_ID) = 1.0000 > 0.05).

[Warning] This channel shows unusually high variability relative to its average value, indicating possible instability. (triggered because coefficient_of_variation(Sample_ID) = 0.5726 > 0.15).

Oil_Pressure_psi,RPM

[Info] Columns 'Oil_Pressure_psi' and 'RPM' are strongly correlated (r = 0.9336), suggesting a possible relationship worth investigating.

[Info] Two channels are strongly correlated, which may indicate a shared physical cause or sensor dependency. (triggered because |r| between Oil_Pressure_psi and RPM = 0.9336 > 0.7).

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[Warning] This row's combination of values across multiple channels does not resemble normal operation, even though no single value looks extreme. (triggered because Row 0 flagged by Isolation Forest).

Conclusion

Analysis identified 15 finding(s) across the dataset. 1 Critical finding(s) require immediate attention. 10 Warning-level finding(s) were detected. 4 informational finding(s) noted. Finding types: correlation pattern, rule match, statistical anomaly, trend pattern.

Recommendations

1. [Critical] Critical: Immediately inspect the cooling system and sensor calibration for the affected channel. (Source: Rule RULE-001: Critical Threshold Breach - Engine Temperature).

2. [Warning] Warning: Schedule a calibration check to rule out sensor drift before the trend approaches operational limits. (Source: Rule RULE-002: Sustained Upward Drift).

3. [Warning] Warning: Cross-check this channel against related sensors for a common root cause. (Source: Rule RULE-003: High Variance Instability).

4. [Warning] Warning: Review this specific timestamp/sample manually — combined-channel anomalies often indicate sensor cross-talk or a transient event. (Source: Rule RULE-006: Multivariate Anomaly Cluster).

5. [Warning] Warning: Review the unusual value(s) in column 'Engine_Temp_C' to determine whether they represent a genuine anomaly or a measurement error. (Source: ZScore detection on column 'Engine_Temp_C').

6. [Warning] Warning: Review the unusual value(s) in column 'Vibration_mm_s' to determine whether they represent a genuine anomaly or a measurement error. (Source: ZScore detection on column 'Vibration_mm_s').

7. [Warning] Warning: Investigate the out-of-range value(s) in column 'Engine_Temp_C' to assess whether they indicate a system issue. (Source: IQR detection on column 'Engine_Temp_C').

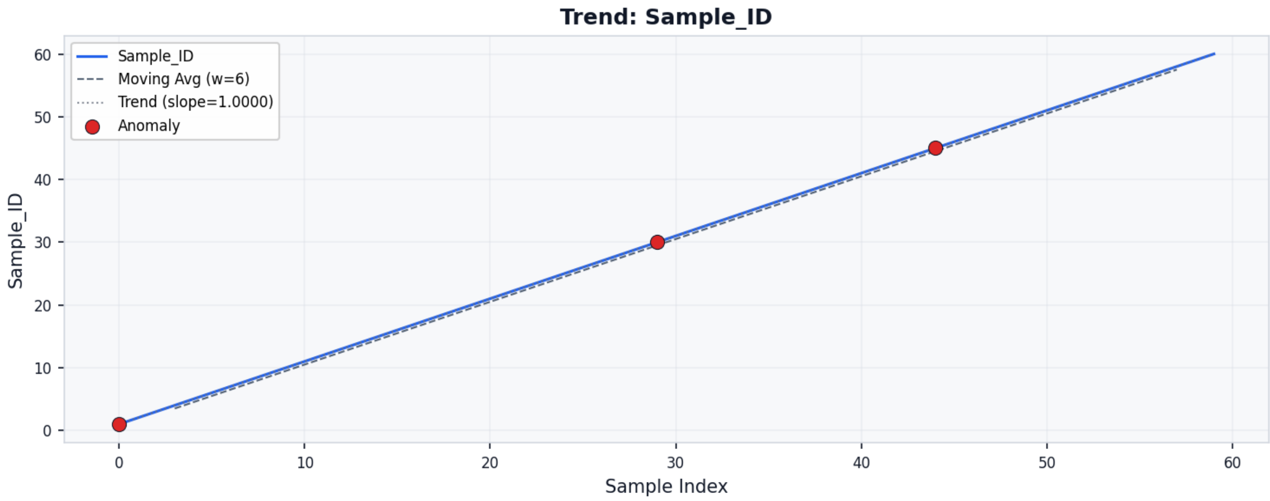
8. [Warning] Warning: Investigate the out-of-range value(s) in column 'Vibration_mm_s' to assess whether they indicate a system issue. (Source: IQR detection on column 'Vibration_mm_s').

9. [Warning] Warning: Examine the flagged rows for unusual multivariate patterns. These may indicate a system-level anomaly not visible in individual columns. (Source: IsolationForest detection on column 'Sample_ID,Engine_Temp_C,Oil_Pressure_psi,RPM,Vibration_mm_s').

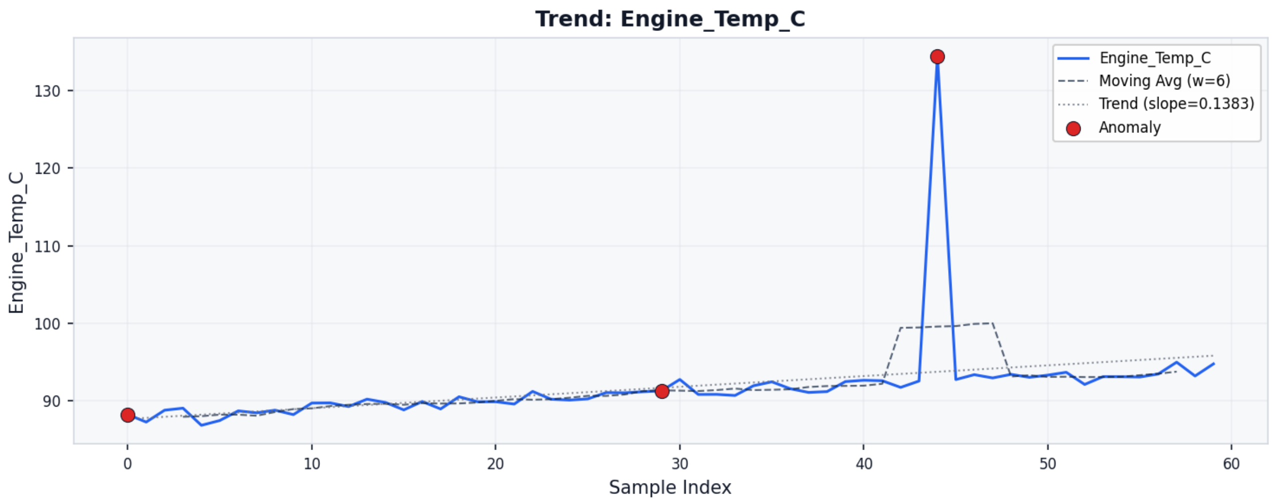
10. [Info] Info: Consider whether one channel could serve as an early-warning proxy for the other. (Source: Rule RULE-005: Strong Correlated Channels).

Graphs

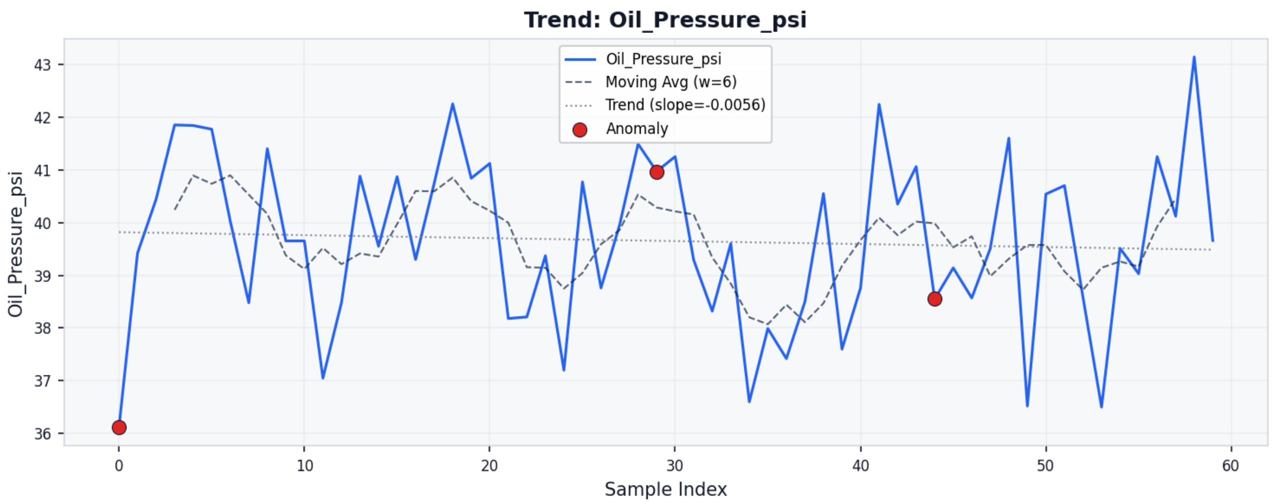
Trend Sample Id



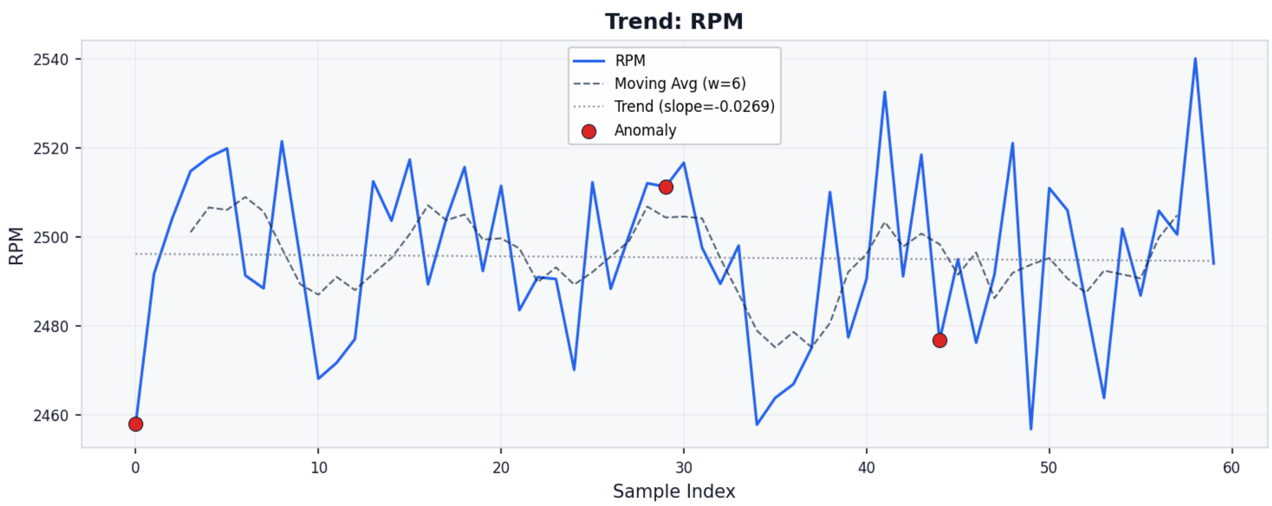
Trend Engine Temp C



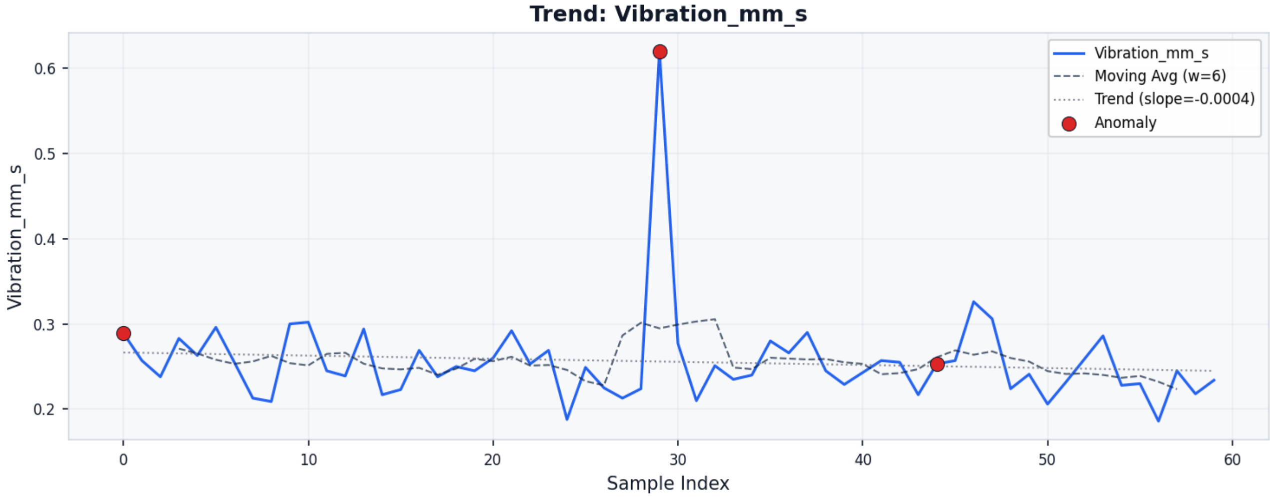
Trend Oil Pressure Psi



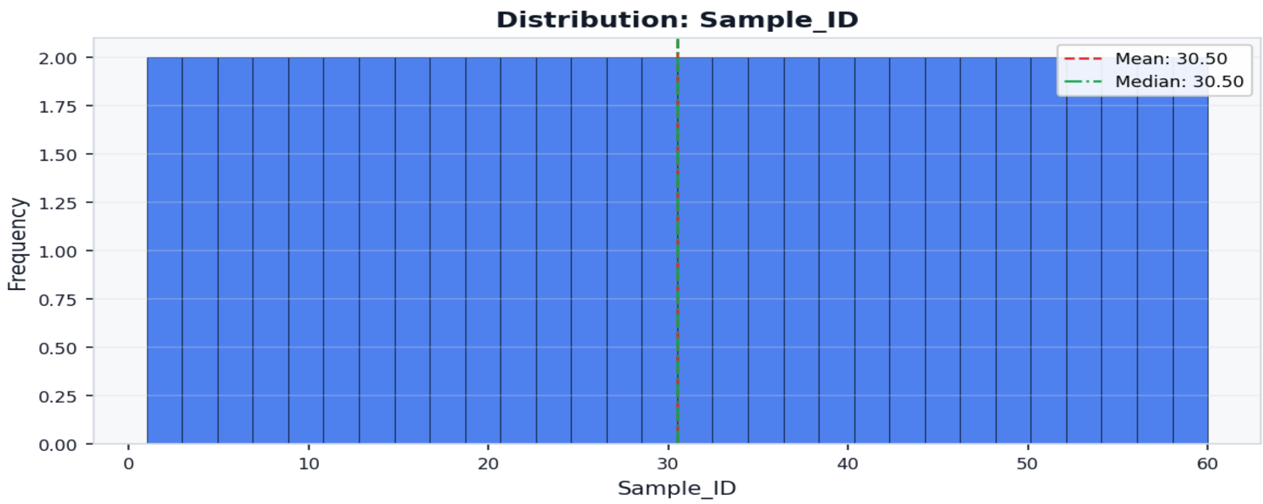
Trend Rpm



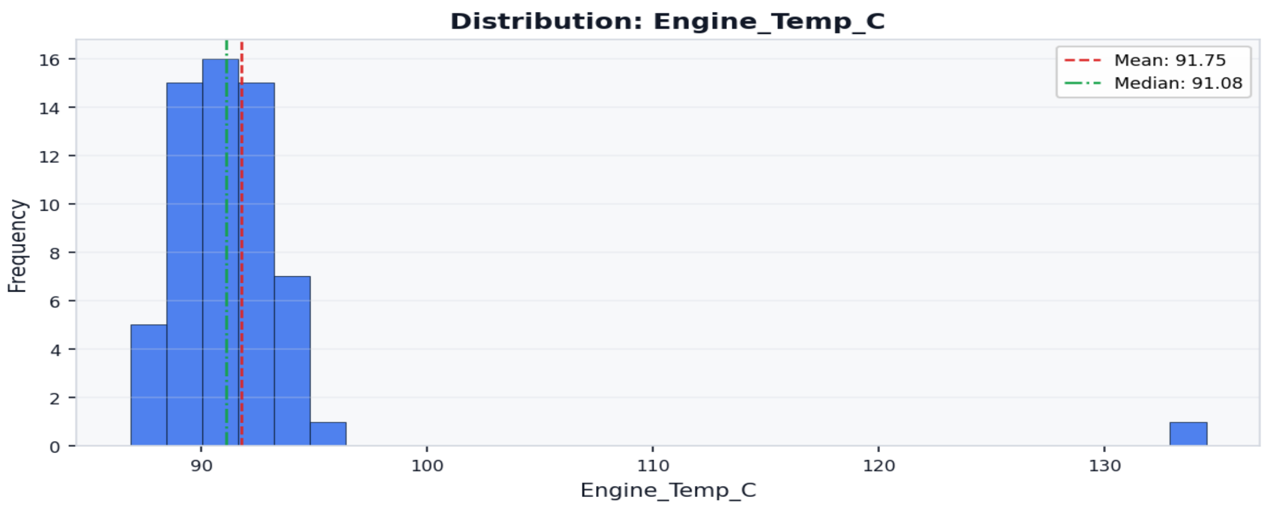
Trend Vibration Mm S



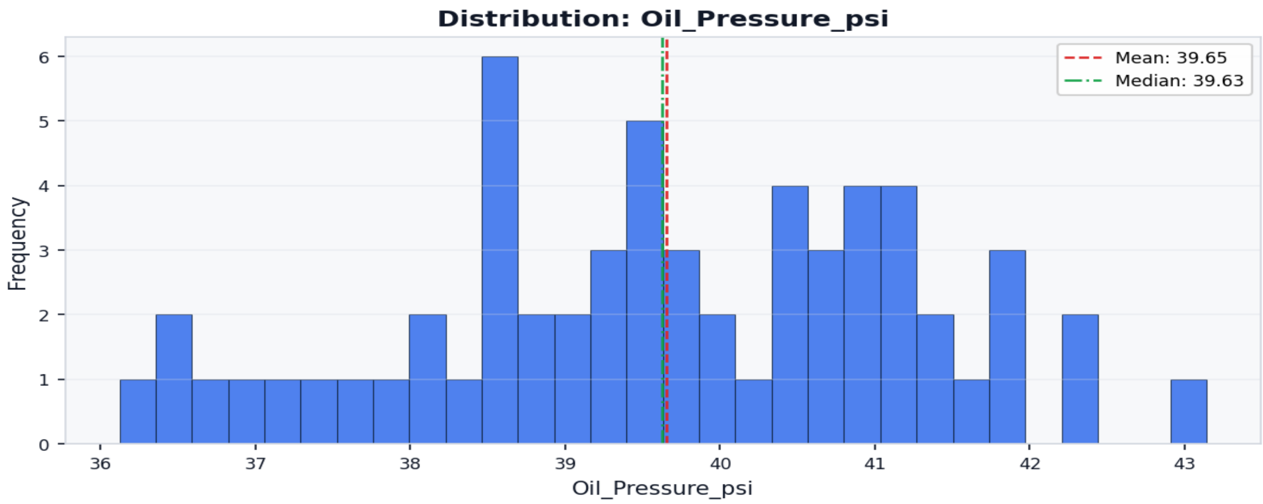
Histogram Sample Id



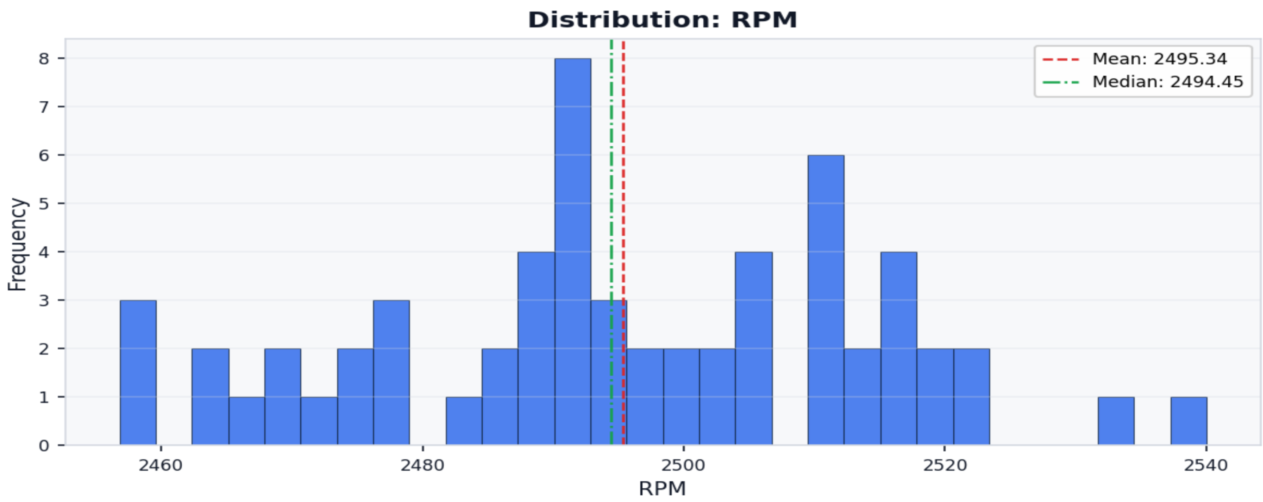
Histogram Engine Temp C



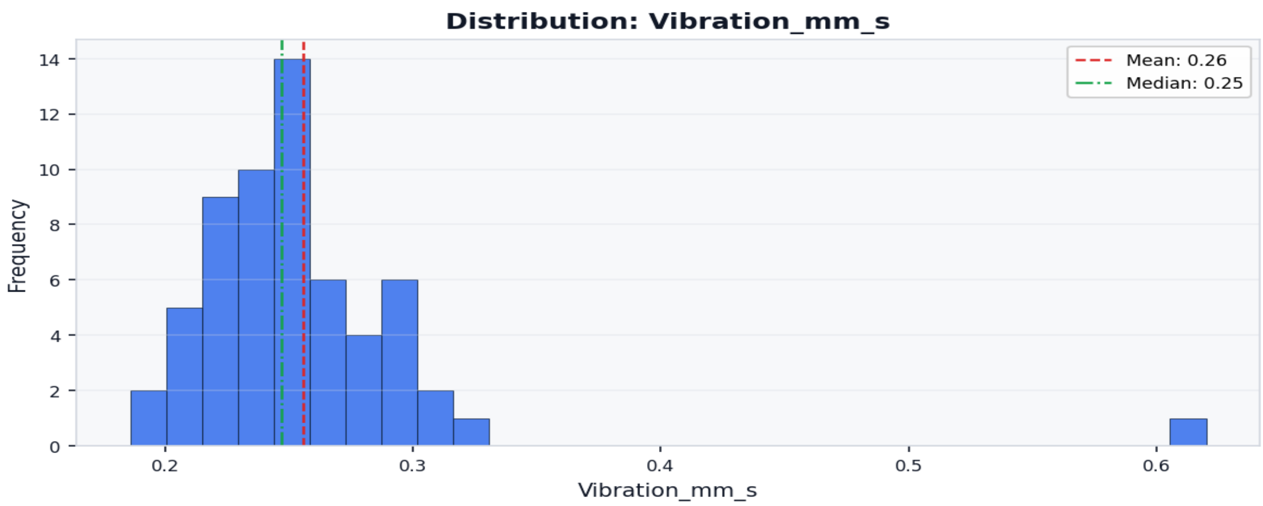
Histogram Oil Pressure Psi



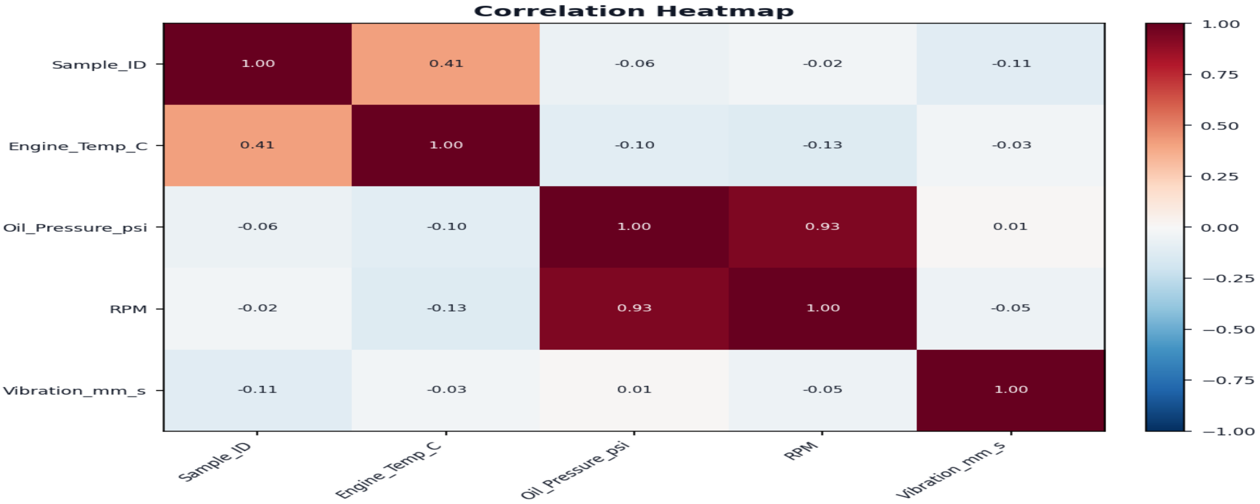
Histogram Rpm



Histogram Vibration Mm S



Correlation Heatmap


Correlation Heatmap
Sample_ID
Engine_Temp_C
Oil_Pressure_psi
RPM
Vibration_mm_s
1.00
0.41
-0.06
-0.02
-0.11
0.41
1.00
-0.10
-0.13
-0.03
-0.06
-0.10
1.00
0.93
0.01
-0.02
-0.13
0.93
1.00
-0.05
-0.11
-0.03
0.01
-0.05
1.00
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00